01.04.02 Applied Mathematics and Computer Science

ITMO

Inverse Calibration of a Continuous-Time SEIRDC Model for Monkey pox Outbreak Dynamics

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Mpox outbreak and motivation

Domain

- Mpox is a zoonotic viral disease with human-to-human transmission
- The 2022 global outbreak exhibited a rapid international spread
- Surveillance data are primarily reported as cumulative cases and deaths

Challenge

- Observations represent aggregated public health indicators
- True epidemiological states (susceptible, exposed, infectious) are unobserved
- Need for a framework linking mechanistic dynamics to reported data

Motivation for this work

- Infer latent epidemic dynamics from surveillance time series
- Estimate key epidemiological parameters
- Provide an interpretable mechanistic reconstruction of outbreak evolution

Dataset

Data source

- Global Mpox surveillance dataset from Our World in Data (OWID)
- Daily country-level reporting aggregated to the global scale
- Public health reporting framework

Temporal coverage

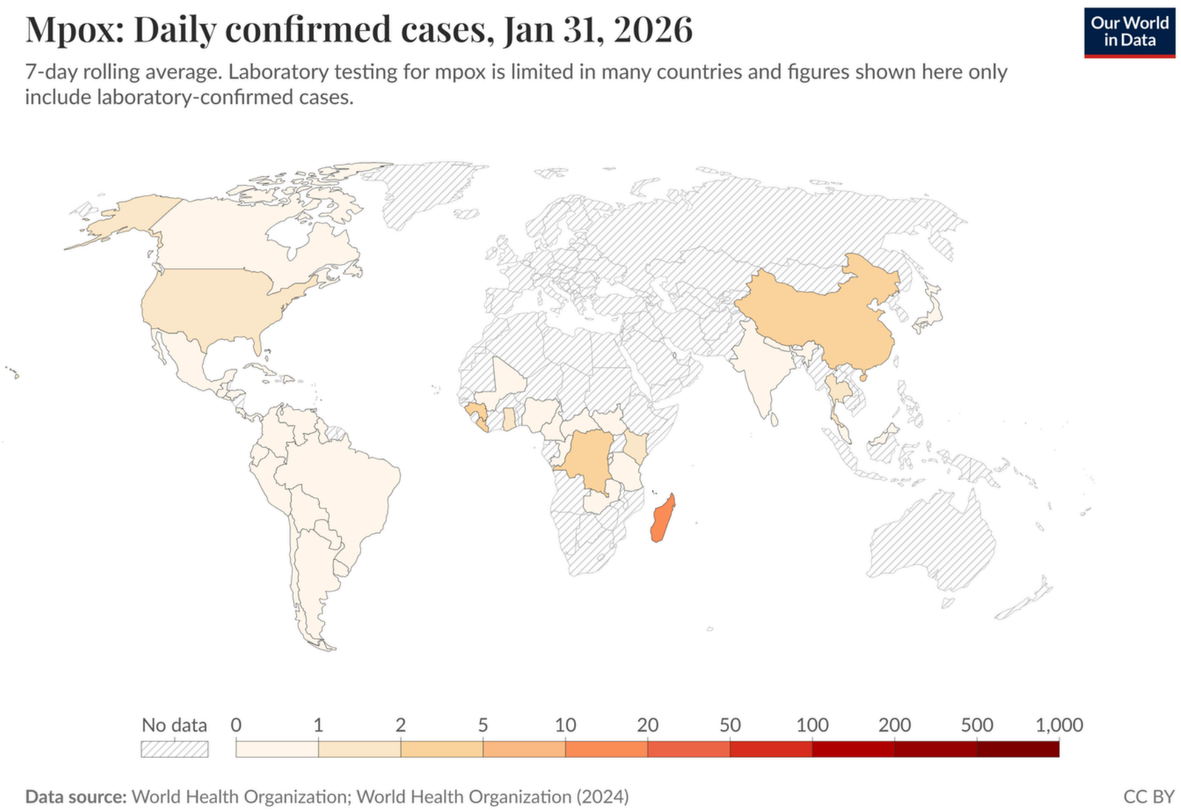
- Observation period: May 2022 – January 2026
- Daily resolution obtained via interpolation of reporting gaps

Observed variables

- Cumulative confirmed cases
- Cumulative reported deaths

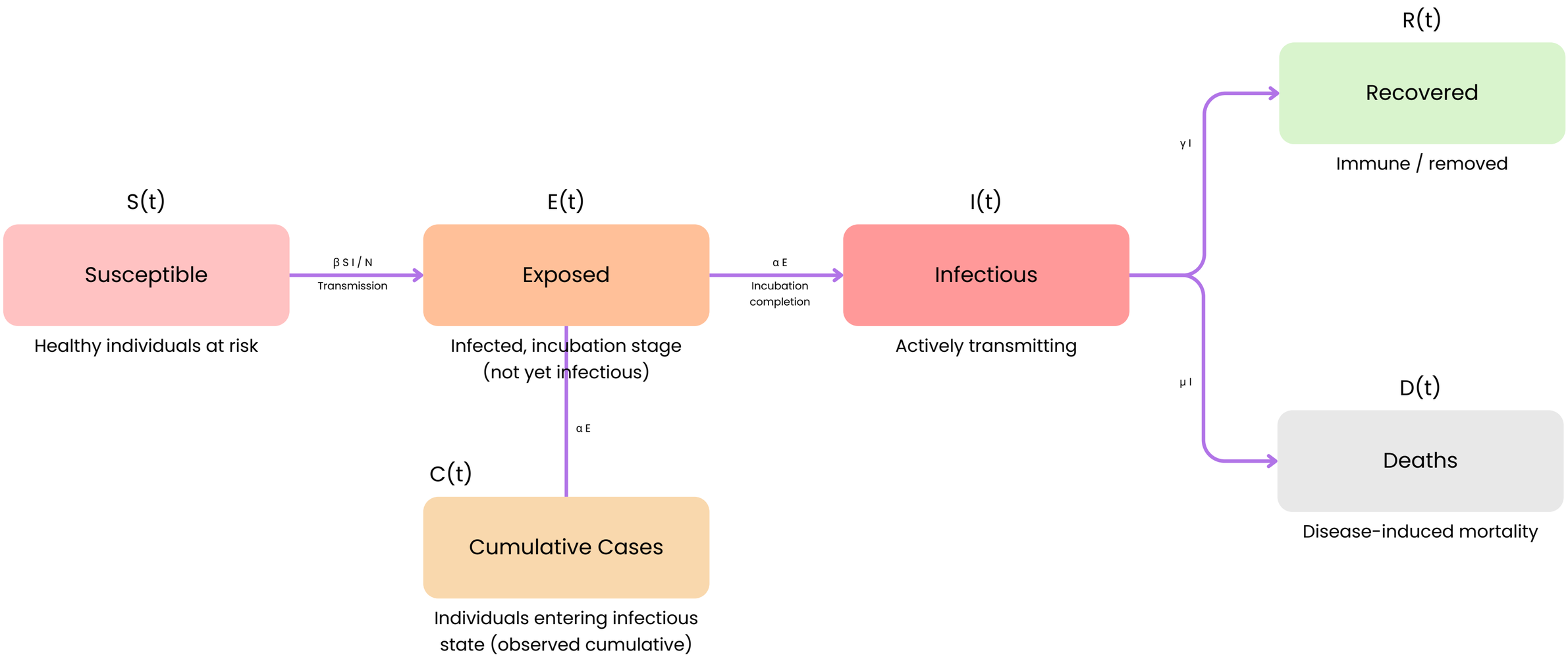
Modeling window

- 7-day smoothed incidence used to detect epidemic peak
- Acute phase selected around peak
- Window: 30 days before → 180 days after



MODEL OVERVIEW

SEIRDC Compartmental Model



Mathematical Formulation

The epidemic dynamics are modeled as a continuous-time compartmental system of ordinary differential equations.

$$dS/dt = -\beta SI/N$$

$$dE/dt = \beta SI/N - \alpha E$$

$$dI/dt = \alpha E - (\gamma + \mu)I$$

$$dR/dt = \gamma I$$

$$dD/dt = \mu I$$

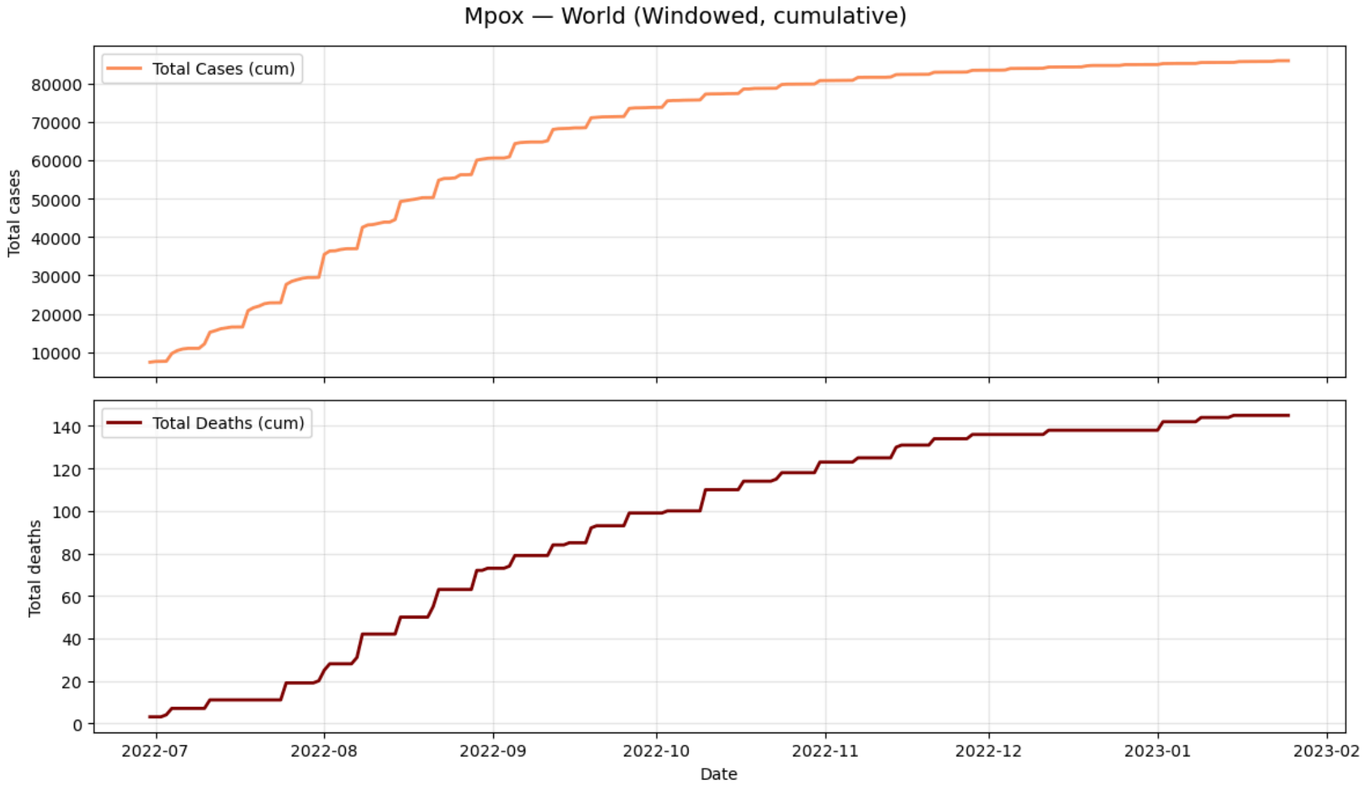
$$dC/dt = \alpha E$$

β – transmission rate
α – incubation rate
γ – recovery rate
μ – mortality rate
N – effective population

The auxiliary state C(t) enables direct calibration of the continuous model to cumulative reported case data.

Data and Observation Structure

- Source: Our World in Data (OWID) Mpox dataset
- Spatial aggregation: Global (World) level
- Temporal resolution: Daily observations
- Variables used:
  - Cumulative confirmed cases
  - Cumulative deaths
- Study period: May 2022 – January 2026



Inverse Problem and Parameter Calibration

Inverse problem

Given observed cumulative trajectories

$$Cobs(t), Dobs(t)$$

estimate parameter vector

$$\Theta = (\beta, \alpha, \gamma, \mu, N, E0)$$

such that model output reproduces observations.

Objective function

Logarithmic mean squared error:

$$L(\Theta) = 1/T \sum (logC(t) - logC_{obs}(t))^2 + 1/T \sum (logD(t) - logD_{obs}(t))^2$$

Optimization strategy

- Global search → Differential Evolution
- Local refinement → L-BFGS-B
- Biologically constrained parameter bounds

Model Calibration Results

Observed vs simulated trajectories

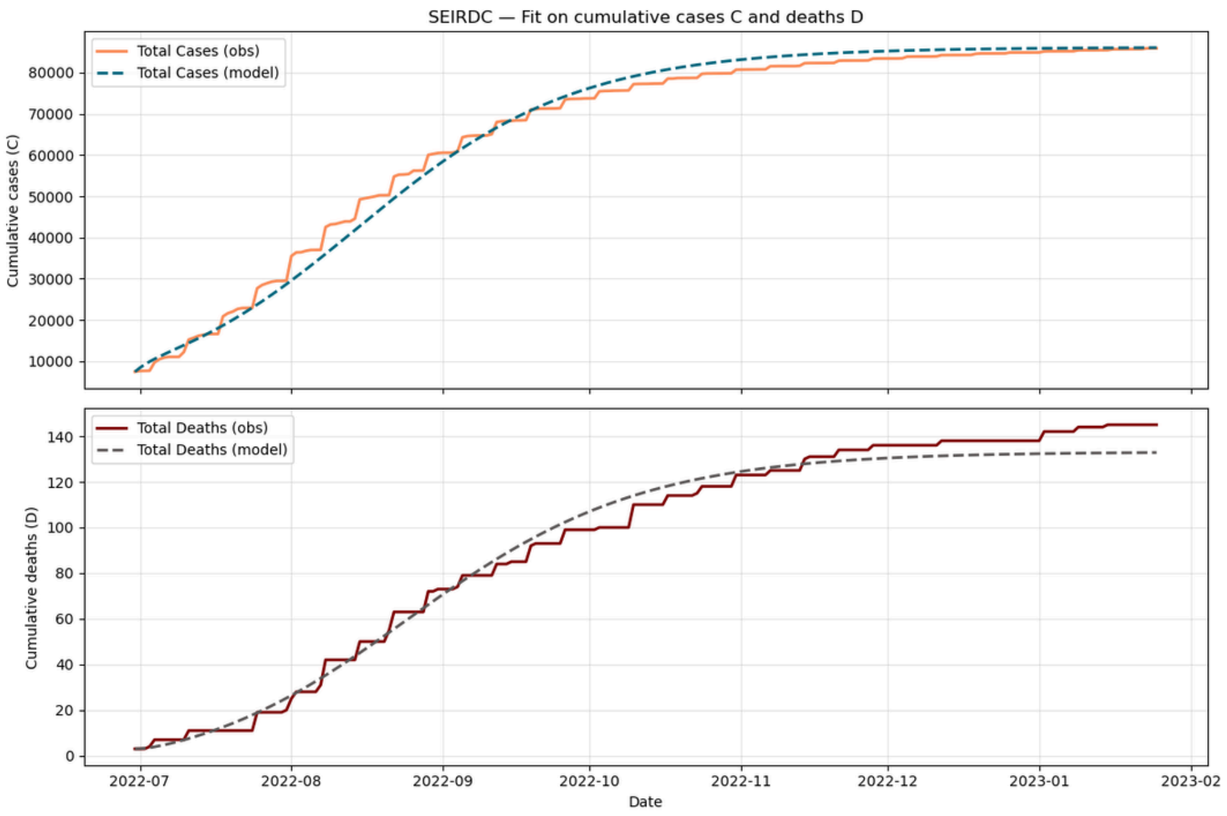
- Continuous-time SEIRDC model fitted to cumulative cases and deaths
- Simultaneous calibration of transmission and outcome dynamics

Goodness of fit

- Log-RMSE (cases): 0.060
- Log-RMSE (deaths): 0.116

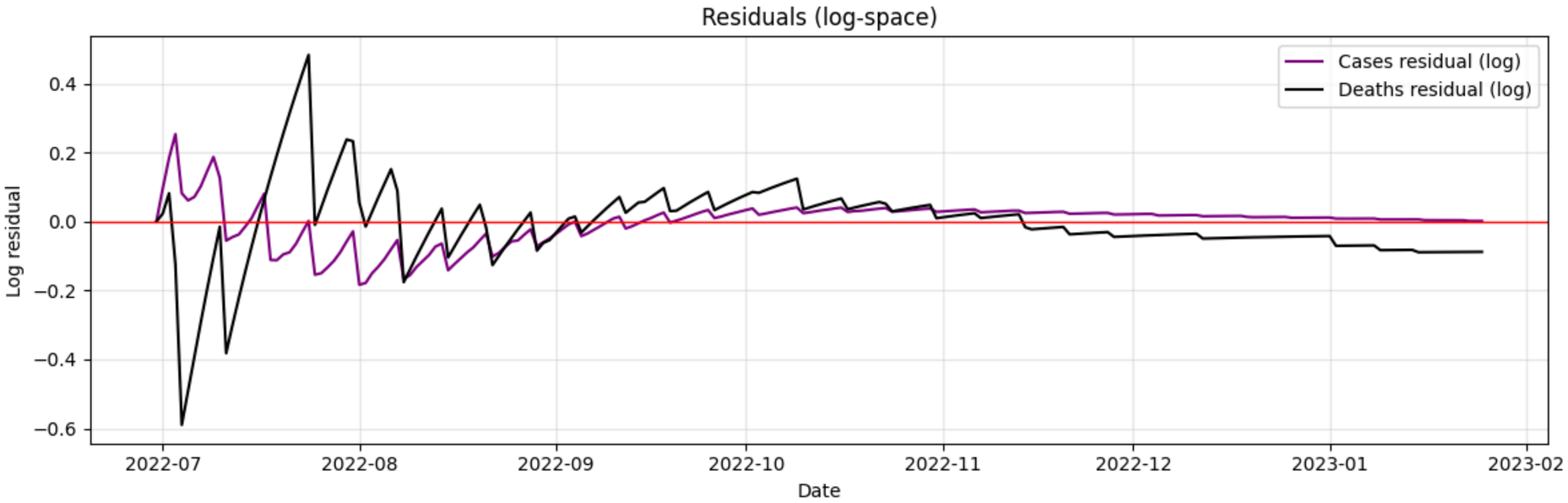
Estimated epidemiological characteristics

- Basic reproduction number R_0≈1.91
- Incubation period ≈ 5.1 days
- Infectious period ≈ 11.2 days
- Case fatality ratio ≈ 0.17%



Residual Analysis

Log-space difference between simulated and observed cumulative trajectories



- Residuals centered around zero, No systematic temporal drift, Larger variance in early outbreak phase
- Model reproduces epidemic growth and plateau regimes
- Remaining deviations attributable to reporting noise

Estimated Epidemiological Parameters

Estimated transmission parameters

- Transmission rate: $\beta = 0.171\ \text{day}^{-1}$
- Incubation rate: $\alpha = 0.196\ \text{day}^{-1}$
- Recovery rate: $\gamma = 0.089\ \text{day}^{-1}$
- Mortality rate: $\mu = 1.47 \times 10^{-4}\ \text{day}^{-1}$

Derived epidemiological characteristics

- Incubation period: ≈ 5.1 days
- Infectious period: ≈ 11.2 days
- Basic reproduction number: $R_0 \approx 1.91$
- Case fatality ratio: ≈ 0.17%

Interpretation

- Moderate transmission potential ($R_0 < 2$)
- Incubation duration consistent with clinical reports
- Low mortality consistent with non-Clade I outbreak dynamics

Discussion and Limitations

Epidemic dynamics

- Single dominant outbreak wave reconstructed
- Growth governed by moderate transmission intensity
- Stabilization explained by susceptible depletion

Modeling assumptions

- Homogeneous population mixing
- Time-invariant transmission parameters
- No behavioral or intervention effects

Methodological contribution

- Latent epidemic states inferred from surveillance time series
- Auxiliary cumulative state enables direct calibration to reported indicators
- Continuous-time formulation supports mechanistic interpretation

Data limitations

- Global aggregation masks regional heterogeneity
- Reporting noise and under-ascertainment
- Lack of active infection observations

Conclusion

This project developed a continuous-time SEIRDC compartmental model to analyze the global Mpox outbreak using publicly reported surveillance data. By explicitly introducing a cumulative infection state, the model enabled direct calibration to observed cumulative cases and deaths while preserving mechanistic epidemic dynamics. The calibrated system successfully reconstructed the dominant outbreak wave and produced epidemiological parameter estimates within biologically plausible ranges. Results demonstrate that latent epidemic states can be inferred reliably even when only aggregated indicators are available. Overall, the study highlights the continued relevance of mechanistic ODE modeling as an interpretable framework for epidemic data analysis.

THANK YOU FOR YOUR TIME!

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